

STX - Seagate Technology | Full Deep Dive

September 21, 2026 | Framework 6.0.2 | NOT HELD | BOTTLENECK - qualified HDD capacity / HAMR leadership | High Conviction

Analytical Base: 47/60 (78%). Valuation: 20x forward four-quarter adjusted operating profit.

Snapshot: USD 877.33 per share on September 21, 2026 close; 226.645 million shares, implying USD 198,842.03 million equity value. Source [STX-6]. Price is a dated header snapshot, not a valuation input in the body.

Active horizon Q3 2027 ceiling: USD 1,194.17/share. Q1 2028: USD 1,356.56/share.

All financial model amounts are USD millions, shares in millions, unless stated. Calendar buckets map to the nearest fiscal quarter: Q3 2026 = FQ1 FY2027, Q4 = FQ2, Q1 2027 = FQ3 and Q2 = FQ4; this mapping repeats. Fiscal year ends around June. July 3 actuals are the June/Q2 bucket, not an extra quarter.

1. Thesis, bottleneck and expanded TAM

Seagate supplies the low-cost, persistent capacity tier of the AI data center. Increasing data creation and retention expands demand even after a GPU is installed. Qualification, precision heads/media fabrication and a limited number of viable HDD manufacturers restrict how quickly demand can be met. The investment thesis is rising exabytes and better economics per drive, not rising PC units.

The bottleneck is commercially durable but not an irreplaceable single-supplier monopoly. HAMR uses localized heating to write smaller magnetic grains; Seagate combines that recording architecture with established manufacturing and customer qualification. Mozaic creates a production lead, while Western Digital and Toshiba remain alternatives. A technical lead earns Core credit; it does not establish exclusivity across future generations. [STX-2, STX-3]

AI video, persistent context and later physical AI broaden the storage opportunity. HDDs hold economical persistent data; HBM/DRAM and SSDs serve active, latency-sensitive computation. The model does not assume HDD directly replaces HBM or that all tokens and video are retained forever. No separate robotics or KV-cache revenue is added on top of the same exabytes.

Competitive check: Western Digital is the most direct large-scale rival; Toshiba remains an HDD producer. NAND suppliers compete at storage-tier boundaries, but current NAND price increases also support HDD economics at the edge. Small photonics companies supply different laser markets and are not a demonstrated substitute for a qualified nearline HDD platform. No credible microcap direct competitor was identified in the filings reviewed. Seagate integrated recording lasers are not automatically a merchant CW-laser revenue opportunity for Sivers. [STX-2, STX-3]

TAM expansion and S-curve

Phase	Economic exposure	Model treatment
2026-27: disclosed ramp	Cloud and enterprise capacity; HAMR takes share within Seagate	No additive HAMR TAM. New drives replace PMR within shipped EB.
2028-29: retention and density	Agentic context, video and wider enterprise adoption	Demand index increases, capped by exabyte supply; no GPU-to-HDD attach ratio assumed.
2030+: physical AI	Robotics, autonomous systems and repeated training data	Embedded only in the modest retention index; no separately priced speculative business.

AH-12 / Plan-B demand bridge: customer identities and supplier wallet shares are confidential. Amazon, Microsoft, Alphabet and Meta form a representative spending basket, not a claim that each is a disclosed Seagate customer. Their latest CY2026 capex figures are \$220B, \$175B, \$200B midpoint and \$137.5B midpoint respectively, totaling \$732.5B [D1-D4]. Definitions differ: Amazon is cash capex, Meta includes finance-lease principal, and Microsoft changed lease classification; the basket is a demand index, not HDD TAM. Memory inflation is not extra physical demand. Calibrate one effective supplier coefficient against the latest actual quarter: $\$2,932\text{M} / (\$732,500\text{M} / 4) = 1.60109\%$. Applying it to the quarterly run-rate basket

reproduces \$2,932M exactly (0.0% calibration error). This coefficient embeds non-basket cloud/enterprise business; individual customer shares are NOT separately knowable. Calibration of one point does not establish forecast accuracy. Each customer is extended at the same analyst-assumed capex growth: +25% in 2027, +20% in 2028, +14% in 2029 and +10% in 2030. Thus basket capex is \$915.625B, \$1,098.750B, \$1,252.575B and \$1,377.833B. Quarterly allocation is one quarter of each annual total, a normalization rather than a forecast of actual quarterly cash spending. Demand EB = 195 anchor EB x (basket capex / \$732.5B) x retained-data intensity. Intensity equals 1.15 / 1.20 in Q3/Q4 2026, 1.25 in 2027-28, 1.30 in 2029 and 1.45 in 2030; these LOW-confidence analyst indices represent accumulating installed-base data. Shipped EB = MIN(demand EB, manufacturing capacity EB). Revenue then equals shipped EB x modeled USD/TB, with no second capex multiplier. Capacity, demand, mix and ASP are separately visible below. This is a calibrated proxy, not a company-disclosed customer model.

2. Latest evidence and financial anchor

Latest two earnings calls reviewed: April 28 and July 28, 2026, including management answers. April emphasized build-to-order visibility, capacity per disk, and modest operating-cost growth. That establishes the operating mechanism; the July financial release and September outlook are the current numerical anchors. [STX-1, STX-3, STX-4]

July management reported 195 data-center EB and 40% HAMR nearline mix. Mozaic 4+ was ramping at two large cloud customers; its next milestone is 50% of HAMR exabytes by calendar year-end, distinct from overall HAMR share. Mozaic 5+ qualification shipments are targeted for late 2027. Fiscal 2027 revenue growth is expected to exceed fiscal 2026 growth of 34%, with sequential profit improvement. [STX-3]

The September 10 management update supports the continuing pricing, areal-density and buyback direction. It reaffirmed the year-end HAMR crossover and described an 80-90% data-center HAMR mix in roughly two years. The model uses 85% by Q2 2028 and 88% in Q3. These are mix assumptions informed by management, not independent booked orders. [STX-5]

Period	Status	Revenue	Gross profit	Operating profit	Source
Q1 2026 / FQ3 FY26	Actual adjusted	3,112.00	1,463.00	1,167.00	STX-4
Q2 2026 / FQ4 FY26	Actual GAAP	3,629.00	1,898.00	1,559.00	STX-1
Q2 2026 / FQ4 FY26	Actual adjusted	3,629.00	1,912.00	1,619.00	STX-1

Financial calibration: Q2 reported data-center revenue is the \$3,629M total less \$697M edge/IoT = \$2,932M. At 195 EB this is \$15.036/TB. For an illustrative three-vector reconstruction of that actual quarter, assume 40% HAMR EB mix and a 4% HAMR price/TB premium: HAMR revenue \$1,200.50M, other data-center revenue \$1,731.50M, edge \$697M. HAMR GM of 61% and edge GM of 35% imply other data-center GM of 54.04% to reproduce \$1,912M adjusted GP. These are calibration proxies, not disclosed product profits. The actual exit-run-rate HAMR mix need not equal quarter-average mix, which limits this decomposition. GAAP-to-adjusted bridge: GP +\$14M and opex -\$46M produce the \$60M OP difference. Ordinary production depreciation remains in gross profit. [STX-1]

Current FQ1 FY2027 guide is revenue \$4,100M +/- \$100M, adjusted opex about \$300M and adjusted operating margin around 50%. The first modeled quarter matches the midpoint: revenue \$4,100M, GP \$2,350M (57.32%), opex \$300M, OP \$2,050M. The split is \$3,375M data center and \$725M edge; 215 shipped EB imply \$15.698/TB. Product GMs are calibrated to the guided total, not presented as management segment guidance. September adjusted EPS guide of \$7.30 +/- \$0.20 is a cross-check, not the valuation input. [STX-1, STX-3]

Balance sheet and capital allocation

Year-end cash was about \$1.7B against \$3.6B debt; FY2026 free cash flow was about \$3.1B. Fiscal 2027 capex is targeted at 4-6% of revenue. Management is retiring more debt and directing later cash generation toward repurchases. The model holds diluted shares constant rather than estimating repurchased shares using an assumed future stock price. Interest and debt repayment affect cash/equity risk but do not enter the framework OP ceiling. [STX-1, STX-3, STX-5]

3. Base score and evidence ledger

Category	Score	Evidence
Core Thesis	18	Hard-to-replicate heads/media and HAMR platform; competitive oligopoly prevents full monopoly credit
Cycle Stage	9	Commercial HAMR ramp and contracts support near-term visibility
TAM Expansion	8	Installed data and AI retention increase exabytes; supply bounds capture
Catalysts	5	Year-end mix crossover, FY2027 ramp, late-2027 next-generation qualification
Macro	9	Cloud spending and persistent data creation; customer construction constraints remain
Alpha	2	Widely covered; analytical edge is capacity/mix decomposition rather than discovery
Risks	-4	Four independent risk classes below
Total	47/60 = 78%	High Conviction

47/60 = 78% (High Conviction). Strong capacity scarcity and an established technology ramp justify the score. Four risk points are counted once each. Price and recent stock performance have no effect on Base. Classification remains BOTTLENECK rather than permanent CHOKEPOINT.

4. Quarterly operating model

Three vectors: HAMR data-center drives, other data-center drives and edge/IoT. For each quarter, capacity-limited shipped EB is split by HAMR share. A 4% analyst-assumed HAMR price/TB premium is normalized so the two prices weight back to the displayed blended price. HAMR revenue = EB x HAMR share x HAMR USD/TB; PMR revenue = EB x (1 - HAMR share) x other USD/TB. With millions of dollars and EB, the conversion factor is one. Each vector has its own GP trajectory. Adjusted R&D and SG&A are explicit percentages; the split of guided opex uses 56%/44%, informed by the reported product-development/marketing mix. No EBITDA substitution.

Revenue vector	Driver and source	Confidence
HAMR data-center drives	STX-3/4/5: qualified ramp and mix milestones; calibrated EB x ASP, analyst supply and premium estimates.	MEDIUM through 2027; LOW 2028-30
Other data-center drives	STX-1/3: actual total less edge, residual EB after HAMR; explicit cannibalization.	MEDIUM through 2027; LOW 2028-30
Edge and IoT	STX-3/5: \$697M actual, constrained stable volume and NAND-related pricing support; modest seasonal extension.	MEDIUM Q3 2026; LOW thereafter

Capacity starts at 215 EB, rises to 285 EB by Q4 2027 and 520 EB by Q4 2030. The quarterly profile represents improving yield and platter density with approximately stable drive units; 2028 shipped capacity rises about 25% on matched quarters. It is a supply estimate, not claimed factory guidance. ASP rises 3%, 3%, 2%, 2%, 1% sequentially through Q4 2027, is flat in 2028, then eases 1% annually in 2029 and 2% in 2030. These assumptions extend current contract momentum without importing NAND inflation. HAMR GP margin reaches 70%, other DC 55%, and edge 43%; cost/mix, not endless price hikes, drive the margin shift. Later margins are LOW confidence. Opex grows from \$300M to \$379M per quarter, with R&D/SG&A percentages falling as revenue scales. Ordinary production depreciation is inside product GM; stock compensation is excluded consistently with the adjusted financial basis and represented in the flat diluted-share assumption.

Physical and commercial drivers

Calendar / fiscal	Demand EB	Capacity EB	Ship EB	USD/TB	New mix (EB)	Main driver / confidence
Q3 2026 / FQ1 FY2027	224.2	215.0	215.0	15.698	45.0%	Guide / MEDIUM
Q4 2026 / FQ2 FY2027	234.0	230.0	230.0	16.169	50.0%	Ramp / MEDIUM
Q1 2027 / FQ3 FY2027	304.7	244.0	244.0	16.654	60.0%	Ramp / MEDIUM
Q2 2027 / FQ4 FY2027	304.7	258.0	258.0	16.987	70.0%	Ramp / MEDIUM
Q3 2027 / FQ1 FY2028	304.7	270.0	270.0	17.326	75.0%	Ramp / MEDIUM
Q4 2027 / FQ2 FY2028	304.7	285.0	285.0	17.500	80.0%	Ramp / MEDIUM
Q1 2028 / FQ3 FY2028	365.6	303.0	303.0	17.500	82.0%	Capacity / LOW
Q2 2028 / FQ4 FY2028	365.6	321.0	321.0	17.500	85.0%	Capacity / LOW
Q3 2028 / FQ1 FY2029	365.6	336.0	336.0	17.500	88.0%	Capacity / LOW
Q4 2028 / FQ2 FY2029	365.6	355.5	355.5	17.500	90.0%	Capacity / LOW
Q1 2029 / FQ3 FY2029	433.5	370.0	370.0	17.456	91.0%	Capacity / LOW
Q2 2029 / FQ4 FY2029	433.5	385.0	385.0	17.412	92.0%	Capacity / LOW
Q3 2029 / FQ1 FY2030	433.5	405.0	405.0	17.368	93.0%	Capacity / LOW
Q4 2029 / FQ2 FY2030	433.5	426.0	426.0	17.325	94.0%	Capacity / LOW
Q1 2030 / FQ3 FY2030	531.9	450.0	450.0	17.237	95.0%	Capacity / LOW
Q2 2030 / FQ4 FY2030	531.9	470.0	470.0	17.151	95.5%	Capacity / LOW
Q3 2030 / FQ1 FY2031	531.9	495.0	495.0	17.064	96.0%	Capacity / LOW
Q4 2030 / FQ2 FY2031	531.9	520.0	520.0	16.978	96.5%	Capacity / LOW

Vector revenue, gross margin and gross profit

For each row: vector gross profit = vector revenue x its own gross margin. The following tables expose the inputs; they are analyst model assumptions unless explicitly identified as reported or guided.

HAMR data-center drives

Quarter	Revenue	Gross margin	Gross profit
Q3 2026	1,551.57	65.00%	1,008.52
Q4 2026	1,895.85	66.00%	1,251.26
Q1 2027	2,476.19	67.00%	1,659.05
Q2 2027	3,103.62	67.50%	2,094.94
Q3 2027	3,542.67	68.00%	2,409.02
Q4 2027	4,020.87	69.00%	2,774.40
Q1 2028	4,378.30	69.20%	3,029.78
Q2 2028	4,802.51	69.50%	3,337.74
Q3 2028	5,198.31	69.80%	3,628.42
Q4 2028	5,620.66	70.00%	3,934.46
Q1 2029	5,897.79	70.00%	4,128.45
Q2 2029	6,186.37	70.00%	4,330.46
Q3 2029	6,559.44	70.00%	4,591.61
Q4 2029	6,953.56	70.00%	4,867.50
Q1 2030	7,383.21	70.00%	5,168.25
Q2 2030	7,711.40	70.00%	5,397.98
Q3 2030	8,121.41	70.00%	5,684.99
Q4 2030	8,531.17	70.00%	5,971.82

Other data-center drives

Quarter	Revenue	Gross margin	Gross profit
Q3 2026	1,823.43	57.66%	1,051.48
Q4 2026	1,822.93	58.00%	1,057.30
Q1 2027	1,587.30	58.00%	920.64
Q2 2027	1,278.96	58.00%	741.80
Q3 2027	1,135.47	57.50%	652.90
Q4 2027	966.56	57.00%	550.94
Q1 2028	924.12	56.50%	522.13
Q2 2028	814.91	56.00%	456.35
Q3 2028	681.60	55.50%	378.29
Q4 2028	600.50	55.00%	330.27
Q1 2029	560.86	55.00%	308.47
Q2 2029	517.26	55.00%	284.49
Q3 2029	474.73	55.00%	261.10
Q4 2029	426.77	55.00%	234.73
Q1 2030	373.64	55.00%	205.50
Q2 2030	349.39	55.00%	192.16
Q3 2030	325.38	55.00%	178.96
Q4 2030	297.52	55.00%	163.64

Edge and IoT

Quarter	Revenue	Gross margin	Gross profit
Q3 2026	725.00	40.00%	290.00
Q4 2026	760.00	41.00%	311.60
Q1 2027	745.00	41.00%	305.45
Q2 2027	765.00	41.50%	317.47
Q3 2027	780.00	42.00%	327.60
Q4 2027	805.00	42.00%	338.10
Q1 2028	785.00	42.00%	329.70
Q2 2028	800.00	42.00%	336.00
Q3 2028	815.00	42.00%	342.30
Q4 2028	840.00	42.00%	352.80
Q1 2029	810.00	42.50%	344.25
Q2 2029	825.00	42.50%	350.62
Q3 2029	840.00	42.50%	357.00
Q4 2029	865.00	42.50%	367.62
Q1 2030	830.00	43.00%	356.90
Q2 2030	845.00	43.00%	363.35
Q3 2030	860.00	43.00%	369.80
Q4 2030	885.00	43.00%	380.55

Combined quarterly earnings

Quarter	Status	Revenue	Gross profit	R&D	SG&A	Other opex	OP	OP margin
Q3 2026	Model	4,100.00	2,350.00	168.00	132.00	0.00	2,050.00	50.00%
Q4 2026	Model	4,478.78	2,620.16	169.68	133.32	0.00	2,317.16	51.74%
Q1 2027	Model	4,808.49	2,885.13	171.36	134.64	0.00	2,579.13	53.64%
Q2 2027	Model	5,147.58	3,154.21	173.04	135.96	0.00	2,845.21	55.27%
Q3 2027	Model	5,458.15	3,389.52	174.72	137.28	0.00	3,077.52	56.38%
Q4 2027	Model	5,792.42	3,663.44	176.40	138.60	0.00	3,348.44	57.81%
Q1 2028	Model	6,087.42	3,881.61	179.20	140.80	0.00	3,561.61	58.51%
Q2 2028	Model	6,417.42	4,130.09	182.00	143.00	0.00	3,805.09	59.29%
Q3 2028	Model	6,694.91	4,349.01	184.80	145.20	0.00	4,019.01	60.03%
Q4 2028	Model	7,061.16	4,617.53	187.60	147.40	0.00	4,282.53	60.65%
Q1 2029	Model	7,268.65	4,781.18	190.40	149.60	0.00	4,441.18	61.10%
Q2 2029	Model	7,528.63	4,965.58	193.20	151.80	0.00	4,620.58	61.37%
Q3 2029	Model	7,874.17	5,209.71	196.00	154.00	0.00	4,859.71	61.72%
Q4 2029	Model	8,245.34	5,469.85	198.80	156.20	0.00	5,114.85	62.03%
Q1 2030	Model	8,586.86	5,730.65	202.16	158.84	0.00	5,369.65	62.53%
Q2 2030	Model	8,905.79	5,953.49	205.52	161.48	0.00	5,586.49	62.73%
Q3 2030	Model	9,306.78	6,233.74	208.88	164.12	0.00	5,860.74	62.97%
Q4 2030	Model	9,713.69	6,516.00	212.24	166.76	0.00	6,137.00	63.18%

Quarterly drivers and confidence

Calendar / fiscal	Status	Revenue	OP margin	OP	Main driver	Confidence
Q3 2026 / FQ1 FY2027	Model	4,100.00	50.00%	2,050.00	Guidance / HAMR 45%	MEDIUM
Q4 2026 / FQ2 FY2027	Model	4,478.78	51.74%	2,317.16	HAMR crossover	MEDIUM
Q1 2027 / FQ3 FY2027	Model	4,808.49	53.64%	2,579.13	Mozaic 4+ scaling	MEDIUM
Q2 2027 / FQ4 FY2027	Model	5,147.58	55.27%	2,845.21	70% HAMR mix	MEDIUM
Q3 2027 / FQ1 FY2028	Model	5,458.15	56.38%	3,077.52	Higher-density volume	MEDIUM
Q4 2027 / FQ2 FY2028	Model	5,792.42	57.81%	3,348.44	Mozaic 5+ qualification	MEDIUM
Q1 2028 / FQ3 FY2028	Model	6,087.42	58.51%	3,561.61	Qualification to ramp	LOW
Q2 2028 / FQ4 FY2028	Model	6,417.42	59.29%	3,805.09	HAMR 85%	LOW
Q3 2028 / FQ1 FY2029	Model	6,694.91	60.03%	4,019.01	Broader adoption	LOW
Q4 2028 / FQ2 FY2029	Model	7,061.16	60.65%	4,282.53	HAMR 90%	LOW
Q1 2029 / FQ3 FY2029	Model	7,268.65	61.10%	4,441.18	Density / retention	LOW
Q2 2029 / FQ4 FY2029	Model	7,528.63	61.37%	4,620.58	Density / retention	LOW
Q3 2029 / FQ1 FY2030	Model	7,874.17	61.72%	4,859.71	Density / retention	LOW
Q4 2029 / FQ2 FY2030	Model	8,245.34	62.03%	5,114.85	Density / retention	LOW
Q1 2030 / FQ3 FY2030	Model	8,586.86	62.53%	5,369.65	Physical-data retention	LOW
Q2 2030 / FQ4 FY2030	Model	8,905.79	62.73%	5,586.49	Density efficiency	LOW
Q3 2030 / FQ1 FY2031	Model	9,306.78	62.97%	5,860.74	Density efficiency	LOW
Q4 2030 / FQ2 FY2031	Model	9,713.69	63.18%	6,137.00	Capacity execution	LOW

Annual trajectory cross-check

The calendar-bucket sums below are a capacity and demand cross-check; quarterly windows remain the live valuation method. Years beyond the latest guidance are model extensions, not company guidance.

Year	Revenue	Operating profit	OP margin	
2027		21,206.64	11,850.30	55.88%
2028		26,260.90	15,668.25	59.66%
2029		30,916.79	19,036.31	61.57%
2030		36,513.12	22,953.89	62.86%

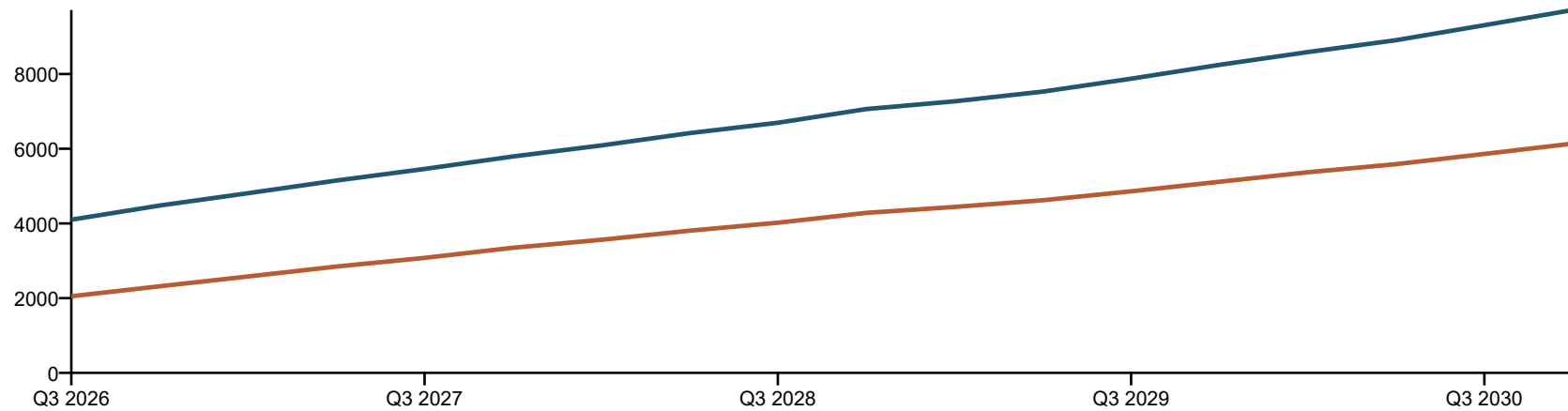
The CY sums below are distinct from fiscal years. FY2027 equals the first four modeled quarters and must exceed $\$12,195\text{M} \times 1.34 = \$16,341.3\text{M}$ to satisfy the current directional outlook. The generated model checks that requirement. The 2028-30 extension depends on technological execution, retention intensity and pricing; no published 2030 HDD TAM or customer-specific allocation is asserted. The four-customer capex basket is deliberately not presented as a storage revenue pool.

Fiscal-year cross-check

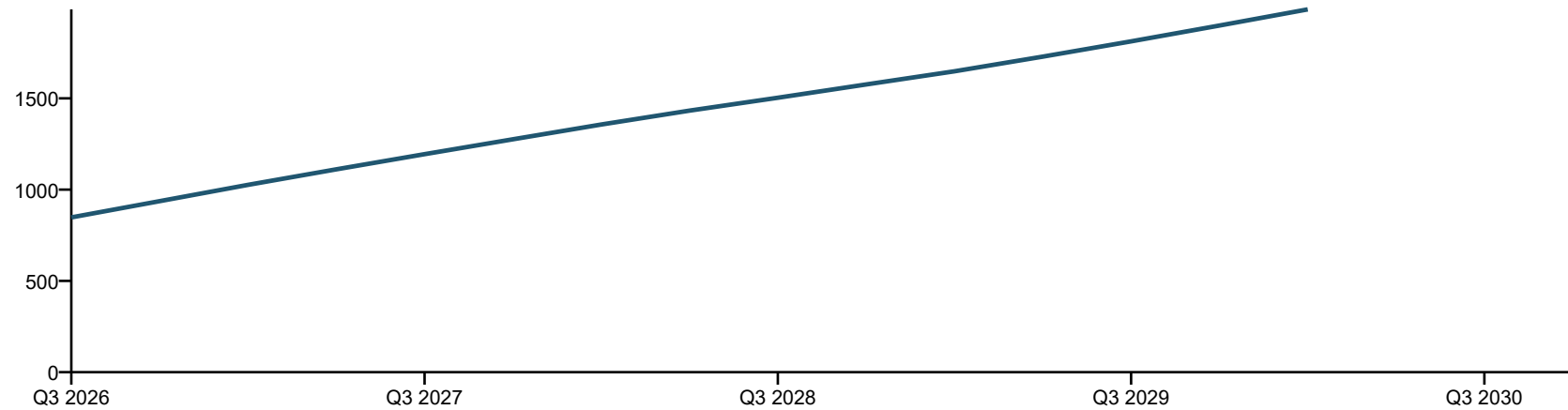
Fiscal year	Exact model window	Revenue	Operating profit
2027	Q3 2026 to Q2 2027	18,534.85	9,791.51
2028	Q3 2027 to Q2 2028	23,755.41	13,792.65
2029	Q3 2028 to Q2 2029	28,553.35	17,363.30
2030	Q3 2029 to Q2 2030	33,612.15	20,930.70

5. Forward four-quarter ceiling and derivations

Revenue (blue), operating profit (orange) - USD M



Forward-four-quarter ceiling - USD/share



Active valuation policy, selected by the portfolio owner on September 21, 2026: 20x forward four-quarter adjusted operating profit for both Seagate and Western Digital. This is a chosen operating-profit multiple under the framework convention, rather than an observed market P/E. The next 12-16 months carry the greatest decision weight, with uncertainty increasing further out. Longer-dated projections remain conditional scenarios. Product execution differences remain in each operating model and Base score.

The framework convention applies the operating-profit multiple directly to diluted equity shares. It is neither a net-income P/E nor an enterprise-value bridge. No separate net-cash addition, tax conversion, probability haircut or price-dependent ranking enters this ceiling. Any probability-weighted revenue is already included once in its operating vector.

Ceiling quarter	Exact earnings window	Forward-4Q OP	Multiple	Diluted shares	Ceiling/share
Q3 2026	Q3 2026 to Q2 2027	9,791.51	20x	231.000	847.75
Q4 2026	Q4 2026 to Q3 2027	10,819.02	20x	231.000	936.71
Q1 2027	Q1 2027 to Q4 2027	11,850.30	20x	231.000	1,026.00
Q2 2027	Q2 2027 to Q1 2028	12,832.78	20x	231.000	1,111.06
Q3 2027	Q3 2027 to Q2 2028	13,792.65	20x	231.000	1,194.17
Q4 2027	Q4 2027 to Q3 2028	14,734.15	20x	231.000	1,275.68
Q1 2028	Q1 2028 to Q4 2028	15,668.25	20x	231.000	1,356.56
Q2 2028	Q2 2028 to Q1 2029	16,547.81	20x	231.000	1,432.71
Q3 2028	Q3 2028 to Q2 2029	17,363.30	20x	231.000	1,503.32
Q4 2028	Q4 2028 to Q3 2029	18,204.00	20x	231.000	1,576.10
Q1 2029	Q1 2029 to Q4 2029	19,036.31	20x	231.000	1,648.17
Q2 2029	Q2 2029 to Q1 2030	19,964.78	20x	231.000	1,728.55
Q3 2029	Q3 2029 to Q2 2030	20,930.70	20x	231.000	1,812.18
Q4 2029	Q4 2029 to Q3 2030	21,931.74	20x	231.000	1,898.85
Q1 2030	Q1 2030 to Q4 2030	22,953.89	20x	231.000	1,987.35

Recognition timing by ceiling

Ceiling quarter	Key inflection entering the forward window
Q3 2026	Guidance / HAMR 45% through 70% HAMR mix
Q4 2026	HAMR crossover through Higher-density volume
Q1 2027	Mozaic 4+ scaling through Mozaic 5+ qualification
Q2 2027	70% HAMR mix through Qualification to ramp
Q3 2027	Higher-density volume through HAMR 85%
Q4 2027	Mozaic 5+ qualification through Broader adoption
Q1 2028	Qualification to ramp through HAMR 90%
Q2 2028	HAMR 85% through Density / retention
Q3 2028	Broader adoption through Density / retention
Q4 2028	HAMR 90% through Density / retention
Q1 2029	Density / retention through Density / retention
Q2 2029	Density / retention through Physical-data retention
Q3 2029	Density / retention through Density efficiency
Q4 2029	Density / retention through Density efficiency
Q1 2030	Physical-data retention through Capacity execution

Complete derivation for every ceiling quarter

Q3 2026: OP window Q3 2026 to Q2 2027: $(2,050.00 + 2,317.16 + 2,579.13 + 2,845.21) \times 20 / 231.000 = \text{USD } 847.75$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q4 2026: OP window Q4 2026 to Q3 2027: $(2,317.16 + 2,579.13 + 2,845.21 + 3,077.52) \times 20 / 231.000 = \text{USD } 936.71$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q1 2027: OP window Q1 2027 to Q4 2027: $(2,579.13 + 2,845.21 + 3,077.52 + 3,348.44) \times 20 / 231.000 = \text{USD } 1,026.00$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q2 2027: OP window Q2 2027 to Q1 2028: $(2,845.21 + 3,077.52 + 3,348.44 + 3,561.61) \times 20 / 231.000 = \text{USD } 1,111.06$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q3 2027: OP window Q3 2027 to Q2 2028: $(3,077.52 + 3,348.44 + 3,561.61 + 3,805.09) \times 20 / 231.000 = \text{USD } 1,194.17$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q4 2027: OP window Q4 2027 to Q3 2028: $(3,348.44 + 3,561.61 + 3,805.09 + 4,019.01) \times 20 / 231.000 = \text{USD } 1,275.68$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q1 2028: OP window Q1 2028 to Q4 2028: $(3,561.61 + 3,805.09 + 4,019.01 + 4,282.53) \times 20 / 231.000 = \text{USD } 1,356.56$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q2 2028: OP window Q2 2028 to Q1 2029: $(3,805.09 + 4,019.01 + 4,282.53 + 4,441.18) \times 20 / 231.000 = \text{USD } 1,432.71$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q3 2028: OP window Q3 2028 to Q2 2029: $(4,019.01 + 4,282.53 + 4,441.18 + 4,620.58) \times 20 / 231.000 = \text{USD } 1,503.32$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q4 2028: OP window Q4 2028 to Q3 2029: $(4,282.53 + 4,441.18 + 4,620.58 + 4,859.71) \times 20 / 231.000 = \text{USD } 1,576.10$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q1 2029: OP window Q1 2029 to Q4 2029: $(4,441.18 + 4,620.58 + 4,859.71 + 5,114.85) \times 20 / 231.000 = \text{USD } 1,648.17$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q2 2029: OP window Q2 2029 to Q1 2030: $(4,620.58 + 4,859.71 + 5,114.85 + 5,369.65) \times 20 / 231.000 = \text{USD } 1,728.55$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q3 2029: OP window Q3 2029 to Q2 2030: $(4,859.71 + 5,114.85 + 5,369.65 + 5,586.49) \times 20 / 231.000 = \text{USD } 1,812.18$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q4 2029: OP window Q4 2029 to Q3 2030: $(5,114.85 + 5,369.65 + 5,586.49 + 5,860.74) \times 20 / 231.000 = \text{USD } 1,898.85$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q1 2030: OP window Q1 2030 to Q4 2030: $(5,369.65 + 5,586.49 + 5,860.74 + 6,137.00) \times 20 / 231.000 = \text{USD } 1,987.35$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

The final three operating quarters support earlier four-quarter windows and have no standalone ceiling. Arithmetic uses unrounded inputs; displayed components are rounded for readability.

Diluted denominator is 231.0M throughout, anchored to the current guidance rather than the 226.645M basic-share snapshot. Convertible dilution is included in that starting estimate. Potential repurchases are assumed only to offset future compensation dilution; no shrinking denominator is used to amplify operating growth. Reconcile with the next filing after note retirement. The sensitivity below isolates denominator risk and is not an alternative operating scenario.

Mechanical share sensitivity

Q1 2028 denominator	Ceiling/share
Current model	1,356.56
5% more shares	1,291.96
10% more shares	1,233.23

6. Cross-book relationships and monitoring

STX <-> WDC: monitor EB growth, HAMR qualification, price/TB, unit discipline and warranty experience as a paired competitive watch. An industry benefit is not automatically share gain for both.

STX <-> SNDK / Kioxia / Samsung / SK hynix / Micron: shared data growth supports tiered storage; NAND inflation can aid HDD price realization in overlapping edge uses, but the prior +23%/+20%/+15% NAND path is NOT applied to HDD ASPs.

STX <-> hyperscaler holdings: spending and deployment delays are common exposures. Optical, networking, compute and storage budgets are complementary pieces of the same infrastructure spend; do not sum their TAMs into Seagate revenue.

7. Scored risks and re-rate tests

Risk class	Score	Failure channel
Temporal demand / customer concentration	-1	Large cloud buyers can defer deployment because of power, permitting or inventory; allocation is not an unconditional shipment guarantee.
Technology and manufacturing execution	-1	HAMR yield, quality, tool-cycle time and qualification delays can limit exabytes or elevate warranty costs.
Structural competition / storage substitution	-1	WDC/Toshiba density progress and improving SSD economics can reduce the pricing premium and retention wallet.
Financial and policy exposure	-1	Debt service, capital spending, export restrictions and manufacturing geography can interrupt cash returns or customer access.

Dated triggers and counters

Checkpoint	Positive evidence	Counter / model action
1. October 2026 result	Q3 revenue near guide and about 50% adjusted OP margin	Below \$4.0B or a margin miss requires a shipment/price/cost bridge, not an automatic extrapolation.
2. December 2026	HAMR >=50% nearline EB; Mozaic 4+ >=50% of HAMR EB	Separate these denominators; reduce advanced mix if either ramp misses.
3. FY2027 / June 2027	Revenue growth above 34%; continued sequential profitability	Rebuild capacity and ASP if growth or pricing stalls; no unapproved Base overwrite.
4. Late 2027 through 2028	Mozaic 5+ qualification then accepted volume shipments	Qualification alone is not full-volume revenue; hold back density steps on delay.
5. Every quarterly filing	Exabytes, costs and cash support the share/opex assumptions	Unexpected dilution or capex above 6% resets the relevant inputs.

Evidence confidence: near-term consolidated guidance is HIGH; inferred product mix and product margins are MEDIUM/LOW; 2028-30 projections are LOW. All modeled quarters are Model, and actuals are separately labeled. No standalone unqualified pipeline is added: no probability adjustment is applied again at the ceiling. If a new speculative adjacency is added later, require a sourced pool, an explicit probability at revenue, timing and cannibalization before including it. Customer-level shares and contract price schedules are unavailable. The model quantifies one base case, not a guarantee of realization. A current

Yahoo high-revenue-estimate table could not be retrieved, so the framework high-estimate cross-check is unavailable; no stale search-snippet estimate is treated as a model floor.

8. Sources and snapshot

- **STX-1** - July 2026 reporting. FY2026 Q4 financial release and GAAP reconciliation. Latest reported results and FQ1 FY2027 guide.
- **STX-2** - August 2026 filing. FY2026 Form 10-K. Competition, customer structure, risks and July 31 basic shares.
- **STX-3** - July 28, 2026. FY2026 Q4 management call. Latest earnings-call operating milestones, EB, edge and guidance.
- **STX-4** - April 28, 2026. FY2026 Q3 official call transcript. Prior management call; actual \$3,112M revenue, \$1,463M adjusted GP and \$1,167M OP checked against the company release.
- **STX-5** - September 10, 2026. Goldman Sachs conference - Seagate management. More recent confirmation of mix, margin direction and capital allocation.
- **STX-6** - September 21, 2026 close. STX dated market snapshot. Header price only; does not determine Base or ceilings.
- **STX-7** - April 28, 2026. FY2026 Q3 results. Prior-quarter numerical anchor.
- **D1** - July 30, 2026. Amazon Q2 management call. Management cash-capex outlook: \$220B in CY2026; increase partly reflects memory cost inflation.
- **D2** - July 29, 2026. Microsoft FY2026 Q4 management call. CY2026 capex about \$175B after lease classification change; physical investment expectations unchanged.
- **D3** - July 22, 2026. Alphabet Q2 management call. CY2026 capex \$195-205B; use \$200B midpoint.
- **D4** - July 29, 2026. Meta Q2 results. CY2026 capex including principal finance-lease payments \$130-145B; use \$137.5B midpoint.

Model snapshot: September 21, 2026. STX. Framework 6.0.2. Base 47/60 (78%). Eighteen modeled operating quarters and fifteen complete forward-four-quarter ceilings. Current full PDF and synchronized Markdown: G:/My Drive/1amInvesting/Artifacts/STX_Seagate_DeepDive.pdf and .md. The companion company worksheet uses this same model and links its ceiling series to the master portfolio.